

Final Report

Customer Classification

The analysis shows that Alron, Aaliyah, and Akil are High-Value Buyers, while Geran is a Low-Value Buyer. Alron has the highest total spending at \$1,079.98, followed by Akil at \$240.00 and Aaliyah at \$159.98. Geran spent \$19.99, placing him in the low-value category.

Total Sales by Category

The Electronics category generated the most revenue at \$999.99, followed by Hardware (\$240.00) and Clothing (\$189.98). Kitchen generated \$49.99, while Home Goods generated the lowest revenue at \$19.99.

Purchase Behavior

The analysis identified that Alron and Aaliyah purchased products from multiple categories, demonstrating broader purchasing behavior than customers with only one order.

Alron was the only customer who purchased from both Electronics and Clothing, purchasing a Laptop and a T-shirt.

The top three highest-spending customers were Alron (\$1,079.98), Akil (\$240.00), and Aaliyah (\$159.98).

Key Business Insights

- Electronics is the strongest revenue-generating category, accounting for the largest portion of sales.
- Alron is the most valuable customer, generating more than four times the spending of the second-highest customer, Akil.
- Customers who purchase across multiple categories may represent opportunities for cross-selling and targeted promotions.
- Geran's low spending suggests an opportunity to encourage additional purchases through promotions or personalized recommendations.
- The presence of customers purchasing from multiple categories demonstrates how customer-level purchase data can be used to identify cross-category purchasing behavior.

Conclusion

Overall, this project demonstrates how Python's lists, tuples, dictionaries, sets, loops, list comprehensions, and sorting can be applied to analyze e-commerce customer data. The analysis provides useful information about customer value, category performance, and purchasing behavior that can support data-driven business decisions.